

Lists, Tuples & Dictionaries

Post-Assessment 3 · 13 questions · 45 points

Name: _____

Date: _____

Score: / 45

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Planet list

4 points

`planets = ["Mercury", "Venus", "Earth", "Mars", "Jupiter", "Saturn"]`. Write what each expression gives.

Expression	Result
<code>planets[2]</code>	
<code>planets[-1]</code>	
<code>planets[3:5]</code>	
<code>len(planets)</code>	

2. Change it

2 points

```
colors = ["red", "green", "blue"]
colors[2] = "pink"
print(colors)
name = "Kim"
name[2] = "d"
```

(a) What does line 3 print?

(1 pt)

(b) What happens on line 5, and why is it different from line 2?

(1 pt)

3. Team Editor

3 points

Write exactly what this program prints.

```
team = ["Ava", "Ben", "Cal"]
team.insert(1, "Dee")
team.append("Eli")
team.remove("Cal")
out = team.pop()
print(team)
print(out)
```

4. Sort it out

3 points

Write exactly what this program prints.

```
scores = [40, 75, 10, 60]
scores.sort()
print(scores)
scores.reverse()
print(scores[-1])
print(75 in scores, 5 not in scores)
```

5. Price report

4 points

Fill in the three blanks so the program finds the **total** and the **smallest** number without using `sum()` or `min()`. Then write what it prints.

```
prices = [4, 9, 2, 7, 5]
total = 0
smallest = prices[0]
for p in _____:
    total = _____
    if _____:
        smallest = p
print(total, smallest)
```

It prints: _____

6. Birthday tuple

3 points

```
birthday = (7, 14)    # (month, day)
month, day = birthday
print(day - month)
```

(a) What does this print?

(1 pt)

(b) What happens if we then run `birthday[1] = 15`?

(1 pt)

(c) Give one other example of data that should be a tuple instead of a list, and say why.

(1 pt)

7. Sentence Sorter

3 points

Write exactly what this program prints.

```
sentence = "dogs chase bouncy red balls"
words = sentence.split()
words.sort()
print(words)
print("-".join(words))
longest = ""
for w in words:
    if len(w) > len(longest):
        longest = w
print(longest)
```

8. Book Profile

4 points

```
book = {"title": "Holes", "pages": 233, "author": "Sachar"}
book["pages"] = 240
book["year"] = 1998
del book["author"]
print(book)
print(book["title"])
print("pages" in book)
```

(a) Write exactly what the program prints.

(3 pts)

(b) What would happen if we added `print(book["rating"])` at the end?

(1 pt)

9. Lunch Vote

4 points

Write exactly what this program prints.

```
votes = {"pizza": 4, "tacos": 7, "salad": 1}
for food, n in votes.items():
    if n >= 4:
        print(food, n)
print(votes.get("sushi", 0))
total = 0
for n in votes.values():
    total = total + n
print(total)
```

10. Word Counter

4 points

```
text = "yes no yes yes maybe no"
counts = {}
for word in text.split():
    if word in counts:
        counts[word] = _____
    else:
        counts[word] = _____
print(counts)
```

(a) Fill in the two blanks.

(2 pts)

(b) Write what the program prints.

(2 pts)

11. List or dictionary?

2 points

For each kind of data, write **list** or **dictionary** and give a short reason.

(a) A phone book of friends' names and phone numbers.

(1 pt)

(b) A to-do list of chores, done in order.

(1 pt)

12. Quiz Bank

4 points

```
quiz = [
    {"q": "2 + 2?", "a": "4", "points": 1},
    {"q": "Capital of France?", "a": "paris", "points": 2},
    {"q": "Largest planet?", "a": "jupiter", "points": 3},
]
for item in quiz:
    if item["points"] >= 2:
        print(item["a"], item["points"])
print(quiz[0]["q"])
```

(a) Write exactly what the program prints.

(3 pts)

(b) Write one line of code that adds a new question: "Opposite of hot?", answer "cold", 1 point. (1 pt)

13. Snack Shop Order

5 points

Use this dictionary of prices:

```
menu = {"chips": 2, "juice": 3, "cookie": 1}
```

Write code that keeps asking `What would you like?` until the user types `done`. Accept messy input (so `" Juice "` counts as juice). Add the price of each item on the menu to a total; for anything else print `Sorry, we don't have that`. At the end print `Your total is $5` (with the real total).

END OF ASSESSMENT